Numerical Methods: Change of Sign Method

Questions 1–10

Question 1. Show that the following equations have a solution in the given interval:

a) $x^3 - 4x - 1 = 0$ for $x \in [2, 3]$
b) $2 + x - e^x = 0$ for $x \in [1, 2]$
c) $x - 4 - 2\ln x = 0$ for $x \in [8, 9]$
d) $1 - x \times 5^x = 0$ for $x \in [0, 1]$

Question 2.

$$\mathrm{f}(x) = \frac{1}{x - 3}$$

a) Find f(2) and f(4)
b) Sketch f(x)
c) Explain why a change in sign of f(x) between $x = 2$ and $x = 4$ is not sufficient to show that there is a root.

Question 3. Draw a sketch for each of these situations for a continuous function:

a) no change of sign, no real roots.
b) no change of sign, one real root.
c) no change of sign, two real roots.

Question 4. Draw a sketch for each of these situations for a discontinuous function:

a) change of sign, no real roots.
b) change of sign, one real root
c) change of sign, two real roots
d) no change of sign, no real roots
e) no change of sign, one real root
f) no change of sign, two real roots

Question 5. Prerequisite knowledge for Q5: Radians

$$f(x) = \frac{1}{2x} - \tan x$$

a) Find f(−1), f(1) and f(2)
b) Explain if there are any roots between $x = -1$ and $x = 1$, and between $x = 1$ and $x = 2$

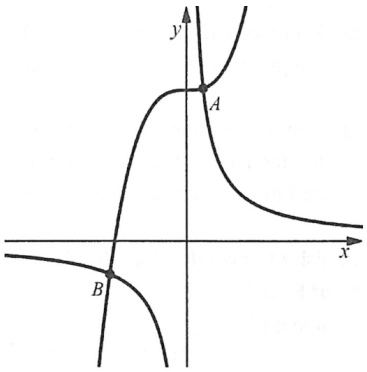
Question 6.

a) Show that the equation $e^x - \frac{1}{x} = 0$ has a root between $x = 0.5$ and $x = 1$
b) Explain whether or not there is a root between $x = -1$ and $x = 1$

Question 7. Sketch the graphs of $y = e^x$ and $y = 4 - x^2$ showing that $e^x = 4 - x^2$ has two solutions.

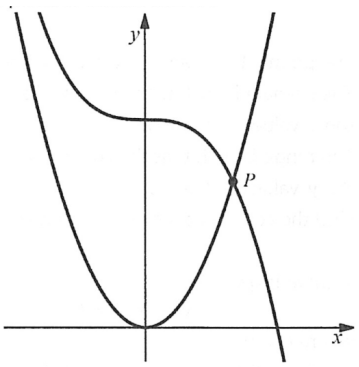
Question 8. Sketch the graphs of $y = \sqrt{x - 1}$ and $y = \frac{1}{2}x$ showing that $\sqrt{x - 1} = \frac{1}{2}x$ has one solution.

Question 9. The diagram shows the graphs of $y = \frac{1}{x}$ and $y = x^3 + 3$



a) Show that the x-coordinates of point A and point B satisfy $x^4 + 3x - 1 = 0$
b) Show that the x-coordinate of point A lies between $x = 0$ and $x = 1$
c) Show that the x-coordinate of point B lies between $x = -2$ and $x = -1$

Question 10. The diagram shows the graphs of $y = x^2$ and $y = -\frac{1}{4}x^3 + 4$ with point of intersection P.



a) Show that the x-coordinate of P is the solution of $x^3 + 4x^2 - 16 = 0$
b) Show that the x-coordinate of point P lies between 1.67855 and 1.67865
c) Given that the x-coordinate of P is α, write down the value of α to 4 decimal places.